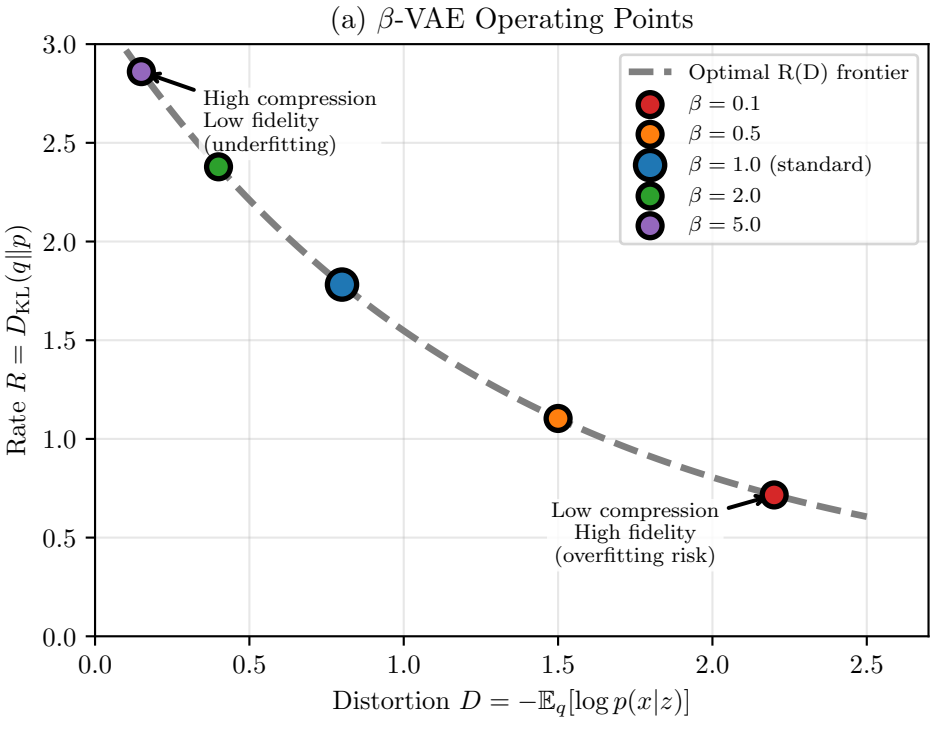

Effect of β on Objective

$$\mathcal{L}_{\beta-\text{VAE}} = \mathbb{E}_q[\log p(x|z)] - \beta \cdot D_{\text{KL}}(q\|p)$$

$\beta < 1$	Prioritize reconstruction	High fidelity
$\beta = 1$	Standard VAE (ELBO)	Balanced
$\beta > 1$	Prioritize compression	Disentanglement

Connection to Information Bottleneck:
β controls how much to compress X into Z